CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instructions & Addressing Modes

Lecture No.- 06



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Recap of Previous Lecture



- Topic: Effective Address
- Topic: Branch Instruction
- Topic: Instruction Cycle
- Topic: Fetch Cycle & Execution Cycle
- Topic: Addressing Modes

Topics to be Covered



- **Topic** Addressing Modes
- **Topic** Types of Addressing Modes
- **Topic** CPU Cycle
- **Topic** CPI



Topic : Implied Mode

The opcode definition itself defines the operand

Opcode	Mode

operation + operand

→ Implied

examples:- INCA → Increment AC



Topic : Immediate Mode

The address field of instruction specifies the operand value

Opcode	Mode	Address

Immediate mode

It is used to use constant value in instruction.

ex:-

MOV R1, #15

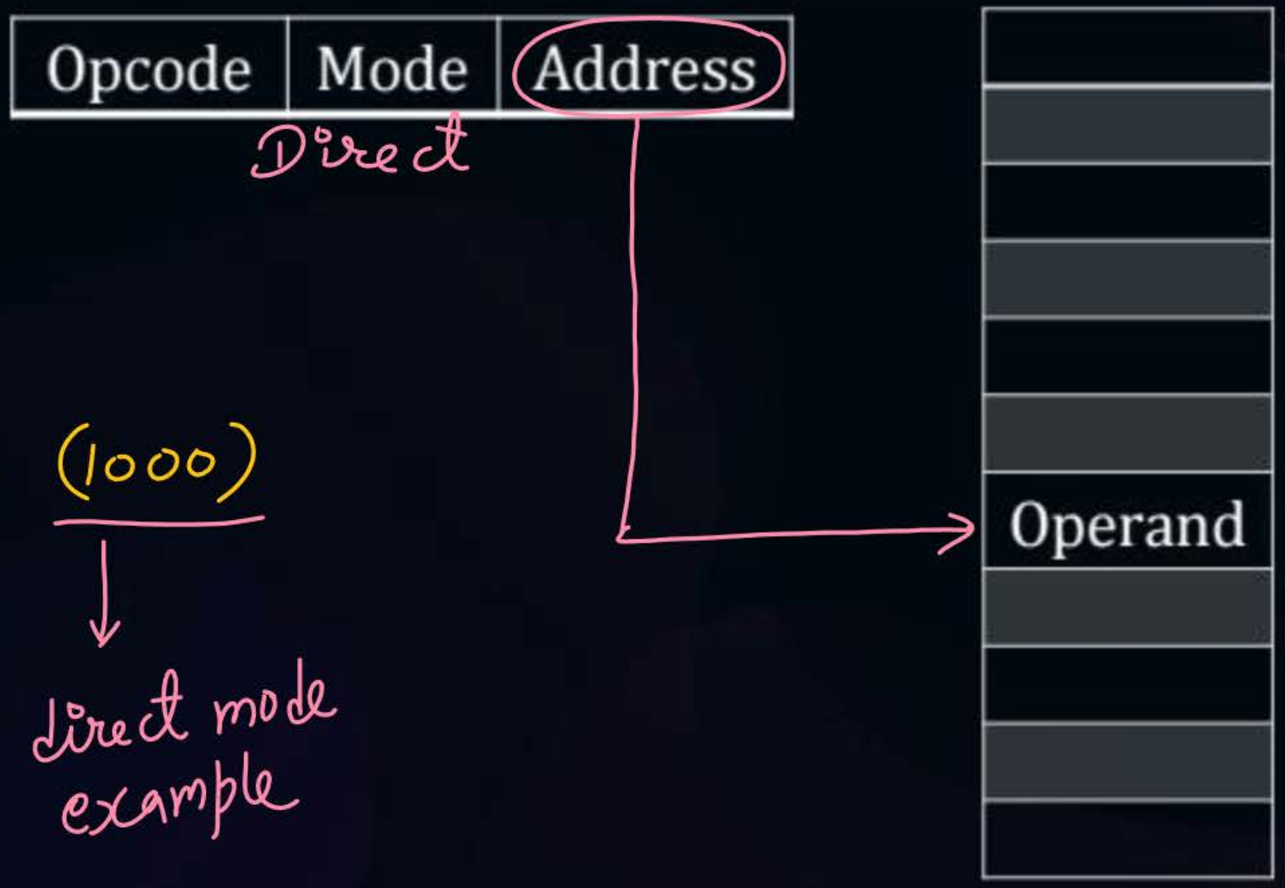
#15 → Immediate Value



Topic : Direct Mode

absolute mode

The address field of instruction specifies the effective address



ex:-

op (1000)

↓

direct mode example

No. of memory references to get operand = 1



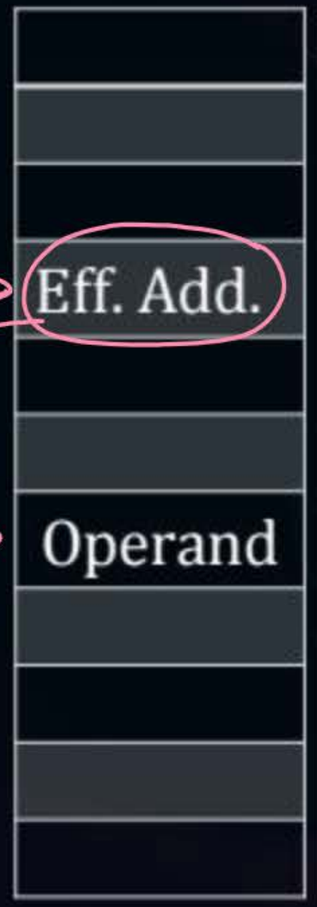
Topic : Indirect Mode

→ used to implement C-language pointers.

The address field of instruction specifies the address of effective address

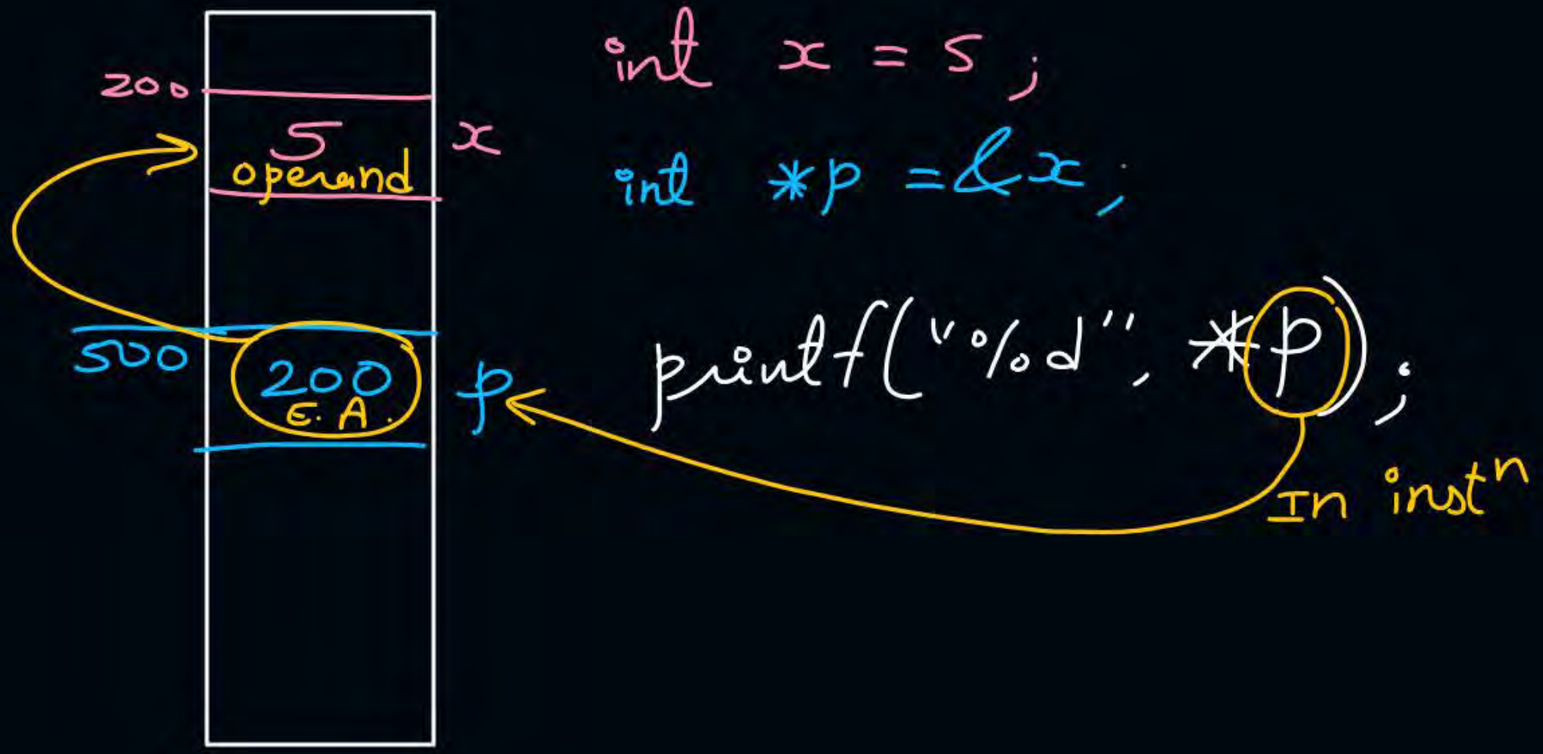
Opcode	Mode	Address

Indirect



ex:- op @1000

No. of mem. references to get operand = 2


200
5
operand
x
500
200
E.A.
p
int x = 5;
int *p = &x;
printf("%d", *p);
In instn



Topic : Register Mode

The address field of instruction specifies a register which holds operand

Opcode	Mode	Address

operand

Register

Ex:- INC R2 → Reg. mode example

ex:- ADD R1, #15 (R1: Reg. mode, #15: Imm. mode)

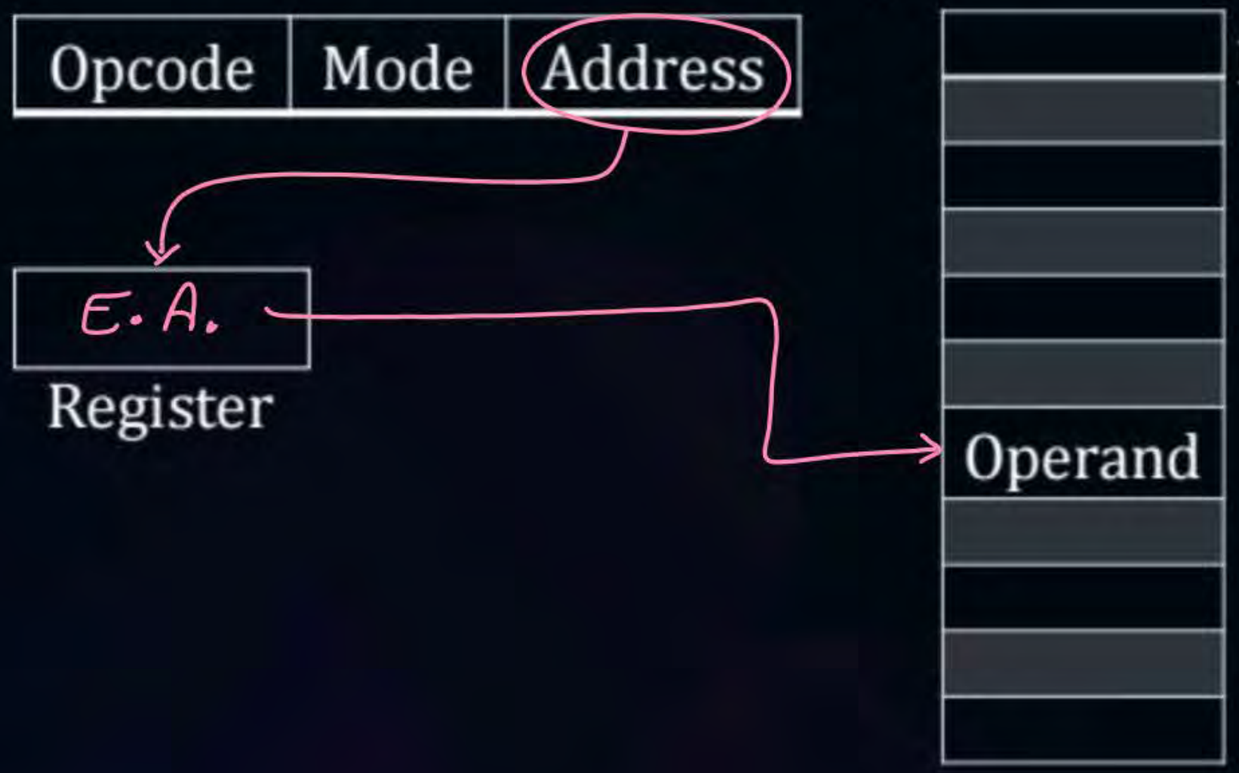
⇓

R1 ← R1 + #15

Topic : Register Indirect Mode

→ used to shorten instⁿ length.
(variable length instⁿ)

The address field of instruction specifies a register which holds ~~operand~~ Effective address.



ex:-

opcode @R1

or

opcode (R1)

or

opcode M[R1

ex:-

Assume a system

Mem. = 4GB = 2^{32}B ⇒ add. = 32 bits

No. of CPU Regs = 64 ⇒ Reg. = 6 bits

Direct mode:-



⇒ (x + y + 32) bits

Reg. Indirect mode:-



⇒ (x + y + 6) bits

Time to get operand

= 1 mem. access time

= (1 Reg. Access time + 1 mem. access time)

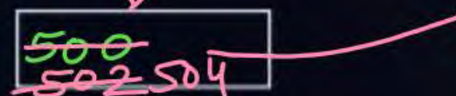
≅ 1 mem. access time

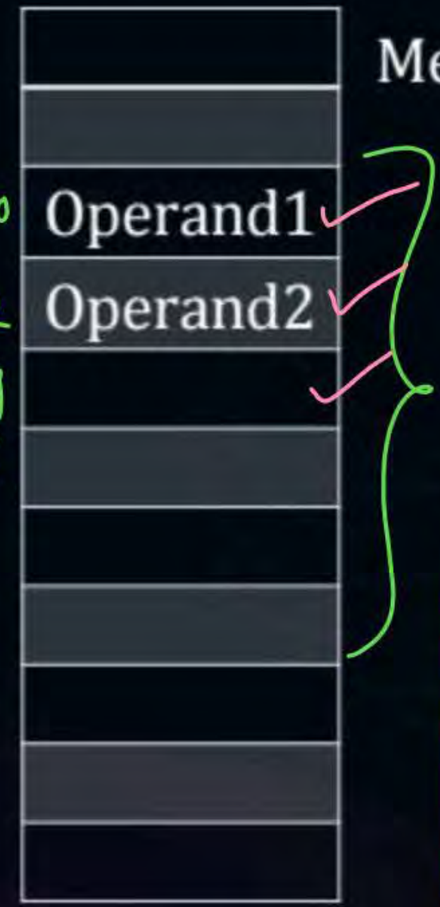
Topic : Autoincrement/Autodecrement Mode

Variant of register indirect mode, in which content of register is automatically incremented or decremented to access a table of content sequentially.

E.A.







→ loop
{ Instⁿ }

In old CPUs :-

Auto inc. mode ⇒ Post inc.

Auto dec. mode ⇒ Pre dec.

Ex:- op (R1)+

op −(R1)

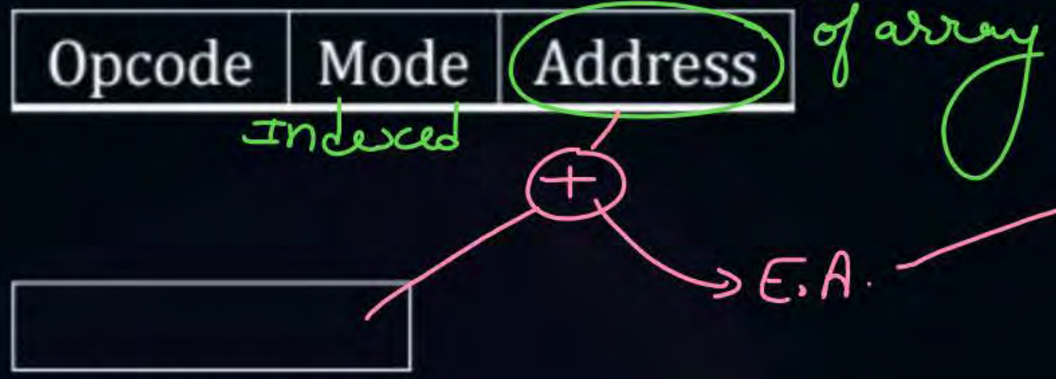


Topic : Indexed Mode

→ used to access specific element of array

↳ Index Reg. mode

Address part of instruction (base address) is added to index register value to get the effective address



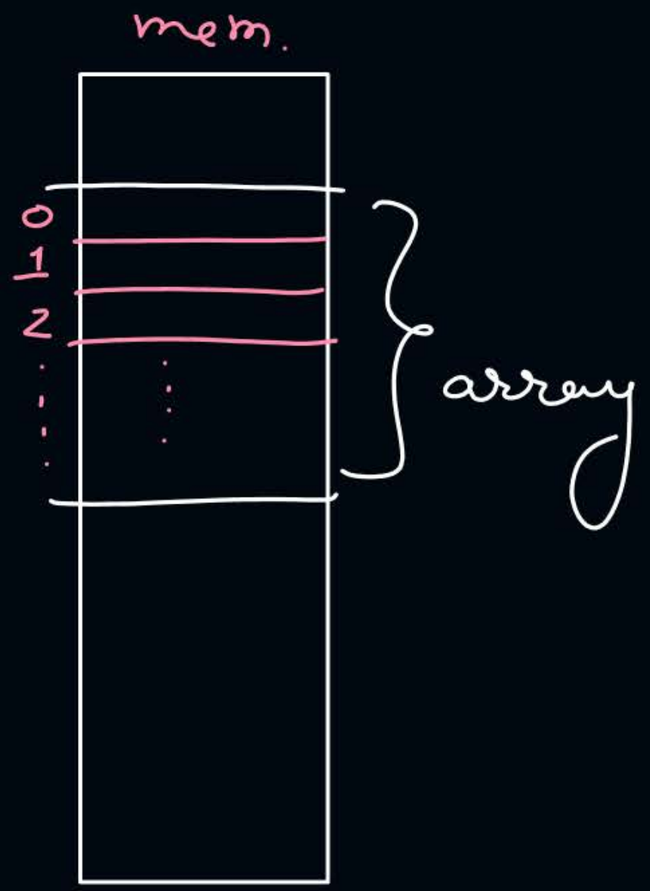


exc :-

INC R1(100)
or
INC 100(R1)

E.A. = R1 + 100

E.A. = Address part of instⁿ + Index Reg. Value



Access array element A[i]

⇓

CPU has to know add. of A[i]

location of A[i] = Base address + (size * i)

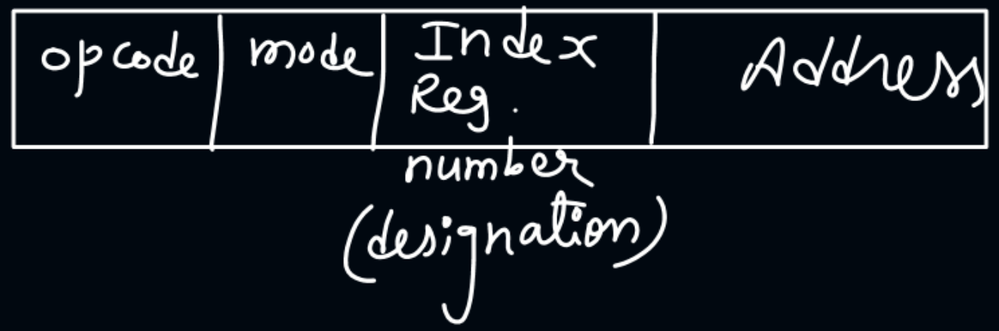
⇓

Inst^ns

✓ Register ← size * i

✓ operation on array element using indexed mode

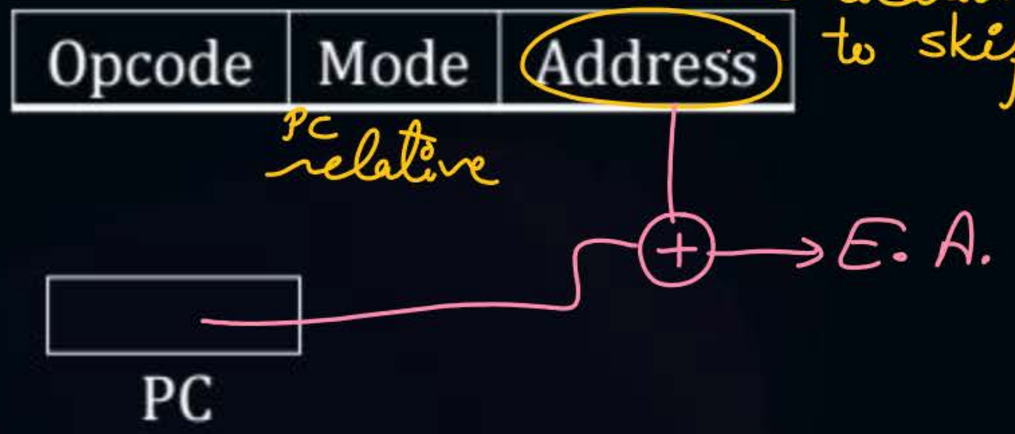
Indexed mode instruct^{n} :-



Topic : PC-Relative Mode / position independent mode

↳ used for intra-segment branching.

Address part of instruction (offset) is added to PC register value to get the effective address





E.A. = Address part of instⁿ + PC Value

Opcode PC (6)

or

opcode 6 (PC)

Assume,

1) CPU fetches I2 instn

PC = 204

2) CPU decodes I2 as branch instn

3)

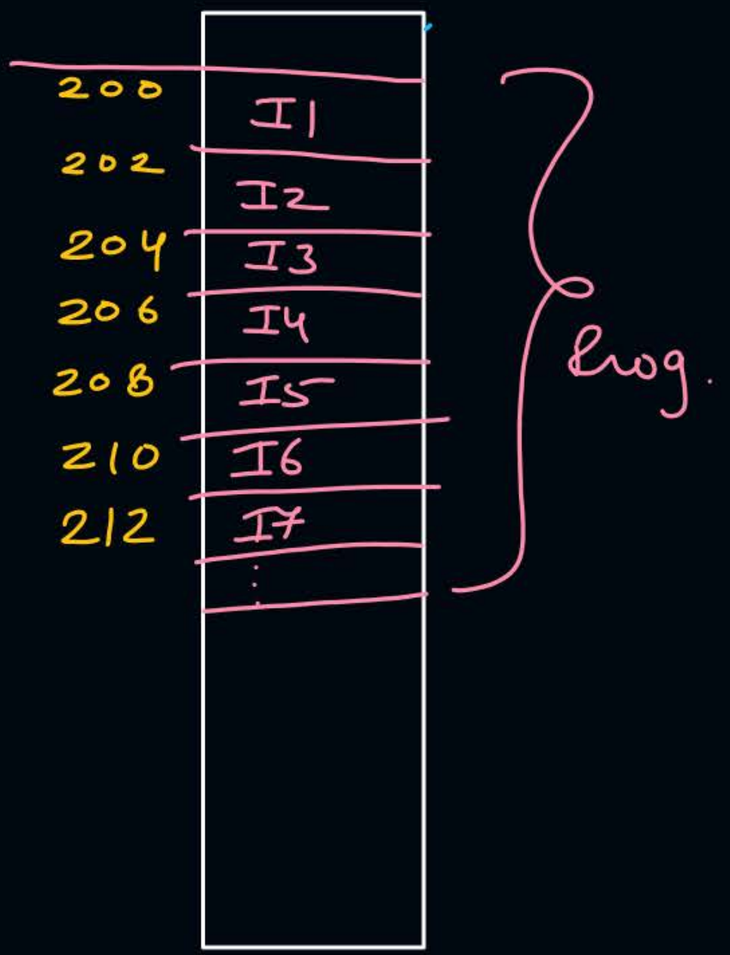
Assume Target of I2 is I6.

Target add. = 210.

Target = Current PC value + no. of relative locations to skip

= 204 + 6

= 210



offset

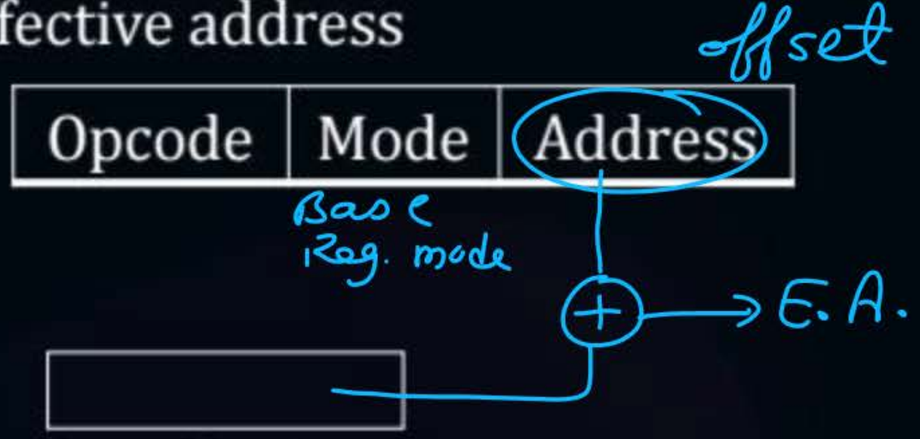
forward jumping ⇒ +ve

backward jumping ⇒ −ve

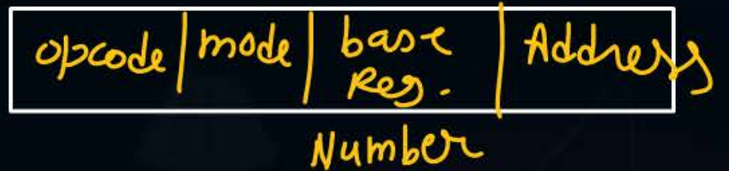


Topic : Base Register Mode → used for inter-segment jumping

Address part of instruction (offset) is added to Base register value to get the effective address







E.A. = Base Reg. Content + Add. part of instⁿ



2 mins Summary

Topic Addressing Modes

Topic Types of Addressing Modes



Happy Learning

THANK - YOU